

TECHNICAL ASSESSMENT REPORT

Linux File System Access Control & Privilege Validation

Target Scope:	/home/researcher2/projects	Date:	June 21, 2026
Classification:	Internal Security Audit	Assigned Analyst:	Muhammad Umar Farhan (SOC Team)

1. EXECUTIVE SUMMARY

A scheduled compliance and privilege posture validation was executed within the storage environments belonging to the corporate Research Division. The objective of this evaluation was to audit existing Discretionary Access Control (DAC) mappings against internal regulatory frameworks and the formal Principle of Least Privilege (PoLP).

Initial baseline observation indicated severe authorization anomalies, specifically globally-writable configurations on structural directories and misconfigured permissions on hidden archival objects. Immediate programmatic remediation was implemented utilizing host-level utilities, effectively containing data leakage vectors and eliminating pathways for localized privilege escalation.

2. ENVIRONMENT DISCOVERY & BASELINE ANALYSIS

Discovery operations were initialized utilizing systemic long listing parameters to expose all directory entries, subdirectories, and standard dot-prefixed hidden administrative objects.

```
Bash Session – Baseline Enumeration

researcher2@sec-ops:~/projects$ ls -la
total 24
drwxr-xr-x 3 researcher2 research 4096 Jun 21 12:00 .
drwxr-xr-x 4 researcher2 research 4096 Jun 21 12:00 ..
drwxr-xr-x 2 researcher2 research 4096 Jun 21 12:00 drafts
-rw-rw-rw- 1 researcher2 research 1024 Jun 21 12:00 project_k.txt
-rw-r--r-- 1 researcher2 research 2048 Jun 21 12:00 project_m.txt
-rwxrwxrwx 1 researcher2 research 512 Jun 21 12:00 .project_x.txt
```

Operational Framework: The integration of the `-l` parameter dictates long form output schema, extracting file system bits, link thresholds, UID/GID bindings, block sizes, and execution timelines. The `-a` argument overrides traditional directory mapping constraints to process dot-prefixed filenames. This approach is standard during security validation because hidden variables frequently mask core environment variables, static backup volumes, or potential indicators of compromise.

3. STRUCTURAL ANALYSIS OF THE DAC PERMISSIONS STRING

Linux operating environments manage security domains via a 10-character bitmask prefix. Evaluating the highly vulnerable `.project_x.txt` permission string (`-rwxrwxrwx`) demonstrates the granular layer assignments:

CHARACTER INDEX	ACCESS LAYER ASSIGNMENT	BIT PATTERN	SECURITY IMPLICATIONS
1	Inode Classification	-	Represents a standard data block (regular file). Directories utilize d .
2 - 4	User Owner Domain	rwX	Full Read/Write/Execute control mapped directly to researcher2 .
5 - 7	Group Domain	rwX	Enables shared project execution for all secondary assets assigned to GID research .
8 - 10	World / Other Domain	rwX	Global authorization structure. Permits any generic system profile to mutate or run the payload.

4. VULNERABILITY REMEDIATION & HARDENING ACTIONS

4.1 World-Writable Mitigation on Production Assets

Corporate compliance architecture forbids global write authorization patterns to preserve raw data integrity. System assessment identified **project_k.txt** as exposing a world-writable state (**-rw-rw-rw-**), which represents an arbitrary data modification vector.

Bash Session – Modifying File Permissions

```
researcher2@sec-ops:~/projects$ chmod go-w project_k.txt
researcher2@sec-ops:~/projects$ ls -l project_k.txt
-rw-r--r-- 1 researcher2 research 1024 Jun 21 12:00 project_k.txt
```

Mechanism: Symbolic masking was used via **go-w**, cleanly stripping the write (**w**) bit flag explicitly from the Group and Others entities while preserving necessary native operations for the owner.

4.2 Absolute Restriction on Hidden Archival Volumes

The archived tracking array **.project_x.txt** displayed a high-risk world-executable and world-writable permission mask (**-rwxrwxrwx**). Mandated criteria required total restriction to a read-only stance for active research peers, while denying any telemetry access to foreign entities.

Bash Session – Absolute Octal Remediation

```
researcher2@sec-ops:~/projects$ chmod 440 .project_x.txt
researcher2@sec-ops:~/projects$ ls -la .project_x.txt
-r--r----- 1 researcher2 research 512 Jun 21 12:00 .project_x.txt
```

Mechanism: Absolute octal bit calculations were applied. Assigning a value of 4 (binary 100) to the User and Group layers provides pure read execution capability, while a trailing 0 completely blocks access for the remaining system entities.

4.3 Isolation of Directory Hierarchies

The subdirectory asset **drafts** handles pre-publication research and requires absolute logical isolation. The baseline configuration allowed universal execution and visibility flags (**drwxr-xr-x**).

```
Bash Session – Directory Boundary Enforcement

researcher2@sec-ops:~/projects$ chmod 700 drafts
researcher2@sec-ops:~/projects$ ls -la | grep drafts
drwx----- 2 researcher2 research 4096 Jun 21 12:00 drafts
```

Mechanism: Executing an octal modification value of **700** limits read, write, and traverse operations strictly to the root owner. Non-authorized profiles executing a `cd` or `ls` within this folder path will hit an immediate host-level access restriction error.

5. SECURITY CONTROL VALIDATION

Following programmatic intervention, a final verification audit was executed. All structural assets within the targeted scope now comply fully with corporate authorization directives.

ASSET IDENTITY	ASSET CLASS	HISTORICAL MASK	REMEDIATED MASK	COMPLIANCE ASSESSMENT
project_k.txt	Standard File	-rw-rw-rw-	-rw-r--r--	Compliant (Least Privilege)
.project_x.txt	Hidden Archive	-rwxrwxrwx	-r--r-----	Compliant (Isolated Archive)
drafts	Directory	drwxr-xr-x	drwx-----	Compliant (Owner Isolated)